11 Synthesis problems, reasoning and starter routines

Sheet 62 Tue 27 Apr. 2027 *Synthesis problems across several areas of the curriculum*

1. Warm-up · Quick Solve $5x - 3 = 2x + 9$.
2. Core · Problem A price rises by 20%, then a fixed discount of 1,000 Ar is applied, giving 11,000 Ar. What was the original price?
3. Core + · Method $f(x) = 2x + 1$ and $g(x) = -x + 7$. Find the point of intersection of the curves, then solve $f(x) < g(x)$.
4. Review · Quick A recipe for 4 people needs 250 g of flour. How much for 6 people?
5. Thread · Quick A shape of area 4 cm^2 is enlarged with scale factor 3. What is the area of the enlargement?

Sheet 63 Thu 29 Apr. 2027 *Proof by contradiction and by cases*

1. Warm-up · Quick What is the parity of $n(n+1)$ for an integer n?
2. Core · Proof by cases Show that $n(n+1)$ is even for every integer n.
3. Core + · Contradiction Show by contradiction that the sum of a rational number r and an irrational number a is irrational.
4. Review · Formula $g(x) = 2x + 5$. Image of -1? Preimage of 11?
5. Thread · Formula → graph Plot the points with abscissae -2 ; -1 ; 0 ; 1 ; 2 on the curve of $f(x) = x^2$; give their ordinates.

Sheet 64 Mon 3 May 2027 *Logical connectives, contrapositive and counter-example in context*

1. Warm-up · Quick Write the negation of "$x > 2$" and then of "$x > 2$ or $x < -1$".
2. Core · Method Write the contrapositive of "if n^2 is even, then n is even".
3. Core + · Counter-example Disprove: "every prime number is odd".
4. Review · True/false True or false? Justify. The perimeter of a square is proportional to the length of its side.
5. Thread · Reading code What does $h(3)$ return? Write $h(x)$ as an expression.

```
def h(x):
    y = x + 2
    return y * y
```